1	2	3	4	5	6	Sum
/20	/20	/20	/20	/20	/20	/100

Name____________________________ January 10, 2020 Friday

SID______________________________ School/Department______________________

University Physics

Final Examination

Kuang Yaming Honors School, Nanjing University

$$\Gamma(\nu)=\int_0^{+\infty} e^{-t}t^{\nu-1}dt,\ \ \Gamma(\nu+1)=\nu\Gamma(\nu),\ \ \Gamma(1)=1,\ \ \Gamma(1/2)=\sqrt{\pi}\ ;\int_{-\infty}^{+\infty} e^{-x^2}dx=\sqrt{\pi}\ .$$

c	h	$\hbar$	e	m_e	m_p	m_n	u	N_A	k_B
3.0×10^{8} m/s	6.63×10^{-34} J·s 1.24×10^{3}eV ·nm/c	1.05×10^{-34} J·s	1.6×10^{-19} C	9.11×10^{-31} kg 0.511MeV/c^2	938.272 MeV/c^2	939.565 MeV/c^2	931.494 MeV/c^2	6.02×10^{23} mol^{-1}	8.617×10^{-5} eV/K

Quantum numbers of three quarks:

Quark symbol	Charge number	Spin	Baryon number	Strangeness
u	2/3	1/2	1/3	0
d	–1/3	1/2	1/3	0
s	–1/3	1/2	1/3	–1

Select five out of the following six problems.

1.(20 pts.) If the electron of a hydrogen atom is replaced by a tau (τ) lepton, a hydrogen tau atom is formed. The rest mass of tau lepton is $m_\tau = 1777.05\ \mathrm{MeV}/c^2$.

(a) Find the reduced mass for the relative motion of the tau (τ) lepton and the proton in the hydrogen tau atom.

(b) By applying the Bohr's quantization rule, find the first Bohr's radius a_0 and the energy levels E_n of the hydrogen tau atom.

2. (20 pts.) The α decay process of a $^{238}\mathrm{U}$ (Uranium-238) nucleus is

$$^{238}\mathrm{U}\rightarrow{}^{234}\mathrm{Th}+{}^{4}\mathrm{He}+Q.。$$

It is known that the decay energy ($Q-$value) of the process is $4.27\ \mathrm{MeV}$. Here we assume that mother nucleus $^{238}\mathrm{U}$ is initially at rest.

(a) Find the kinetic energy of the emitting α particle ($^{4}\mathrm{He}$ (Helium-4) nucleus).

(b) Find the kinetic energy of the recoiling daughter $^{234}\mathrm{Th}$ (Thorium-234) nucleus.

3. (20 pts.) For each of the following reactions, identify it as allowed or forbidden. If it is forbidden, state at least one conservation law that is violated. If it is allowed, state the main interaction the reaction is mediated.

(a) $\mathrm{p}+\mathrm{p} \to \mathrm{p}+\mathrm{p}+\mathrm{K}^{+}+\mathrm{K}^{-}$

(b) $\mathrm{p}+\mathrm{K}^{-} \to \Sigma^{+}+\mathrm{n}+\pi^{+}+\pi^{-}$

(c) $\pi^{+}+\mathrm{p} \to \mathrm{K}^{+}+\Sigma^{+}$

(d) $\bar{\nu}_{\mu}+\mathrm{p} \to \mu^{+}+\mathrm{n}$

(e) $\mathrm{p}+\mathrm{p} \to \mathrm{p}+\mathrm{n}+\mathrm{K}^{+}$

(f) $\mathrm{K}^{+} \to \pi^{+}+\pi^{0}$

(g) $\mathrm{K}^{+} \to \mu^{+}+\nu_{\mu}$

allowed/forbidden	violated conservation law/main interaction
(a)	
(b)	
(c)	
(d)	
(e)	
(f)	
(g)	

The quark combinations of relevant hadrons:

$\mathrm{p}-\mathrm{uud}$ $\quad \mathrm{n}-\mathrm{udd}$ $\quad \Sigma^{+}-\mathrm{uus}$ $\quad \mathrm{K}^{+}-\mathrm{u}\bar{\mathrm{s}}$ $\quad \mathrm{K}^{-}-\mathrm{s}\bar{\mathrm{u}}$ $\quad \pi^{+}-\mathrm{u}\bar{\mathrm{d}}$ $\quad \pi^{-}-\mathrm{d}\bar{\mathrm{u}}$

4. (20 pts.) The electronic stationary wave function of a hydrogen atom at its ground state can be written in the form

$$\psi(r) = Ce^{-r/a_0}$$

where r is the radial distance in the spherical coordinates, a_0 the Bohr radius and C the normalization coefficient to be determined.

(a) Determine the normalization coefficient C in terms of a_0.

(b) Find the corresponding radial probability density $w(r)$.

(c) Calculate the most probable radius $r_{\max}$ and the average radius $\langle r \rangle$ of the electron.

5. (20 pts.) A particle of mass m moves in a one-dimensional infinite potential well

$$U(x) = \begin{cases} 0, & 0 < x < a, \\ +\infty, & x \le 0 \text{ or } x \ge a, \end{cases}$$

where $a > 0$ is the width of the well. There is a normalized wave function

$$\psi(x,t) = \begin{cases} \frac{3}{5}\sqrt{\frac{2}{a}} \sin(k_1 x) e^{-i\omega_1 t} + A\sqrt{\frac{2}{a}} \sin(k_2 x) e^{-i\omega_2 t}, & 0 < x < a, \\ 0, & x \le 0 \text{ or } x \ge a, \end{cases}$$

where $k_1 = \pi/a$, $k_2 = 2\pi/a$, $\omega_1 = \hbar k_1^2/(2m)$, $\omega_2 = \hbar k_2^2/(2m)$ and A is a constant to be determined.

(a) Verify that the wave function satisfies the Schrodinger equation and the boundary conditions for the potential well.

(b) Determine the constant A.

(c) Find the corresponding probability density $\rho(x,t)$ of the particle.

(d) Find the average position $\langle x \rangle$ of the particle as the function of time t.

6.(20 pts.) Consider a gas of N extremely relativistic spin-$1/2$ fermions that are confined in a volume V. The energy-momentum relation of the fermions is $\varepsilon = pc$.

(a) Find the density of states (i.e., the number of quantum states per unit energy interval) of the system.

(b) Find the Fermi energy E_F of the system.

(c) Show that the total energy of the system at $T = 0\,\mathrm{K}$ is $(3/4)NE_F$.